A Phenomenological Suppression of Small-Scale Power at $k_c \approx 0.75\,h\,\mathrm{Mpc}^{-1}$: An Exploratory Test of DGP-Inspired Leakage

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Abstract

We investigate whether a suppression of small-scale matter power at a characteristic comoving wavenumber $k_c \approx 0.75\,h\,\mathrm{Mpc}^{-1}$ can simultaneously alleviate several observational tensions. Motivated by braneworld scenarios in which gravitational modes may leak into an extra dimension, we implement a multiplicative transfer function $T(k) = \exp[-(k/k_c)^p]$ (fiducial $p = 2.0$) in the linear matter power spectrum.

Applied to current data, the modification improves the fit to the ACT DR6 and SPT-3G high-ℓ CMB damping tail by $\Delta\chi^2 \approx -10.2$ and reduces the inferred clustering amplitude from $S_8 = 0.830$ to $S_8 = 0.768$, closer to DES Y6 measurements. It also increases the predicted abundance of massive halos at $z \gtrsim 10$ by ~ 35–50%, consistent in direction with JWST observations. The preferred scale lies in the same regime where several small-scale structure tensions have historically been discussed. The value of k_c is determined empirically; the DGP-inspired interpretation is presented as one possible physical motivation rather than a unique derivation. We discuss shape discrimination, Lyman-α consistency, degeneracies, and concrete observational tests for future surveys.

1 Introduction

The standard ΛCDM model provides an excellent description of the universe on large scales, yet several tensions remain at smaller scales. These include excess damping in the CMB temperature power spectrum at high multipoles, a lower clustering amplitude S_8 from weak-lensing surveys compared to Planck, and an overabundance of compact galaxies at high redshift reported by JWST. Additionally, long-standing small-scale structure problems in cold dark matter point to the need for some form of suppression of power on comoving scales around $k \sim 0.5$–$2\,h\,\mathrm{Mpc}^{-1}$.

Several of these tensions probe partially overlapping comoving scales within the approximate range $k \sim 0.5$–$2\,h\,\mathrm{Mpc}^{-1}$. This motivates investigation of whether a single scale-dependent suppression mechanism may contribute to multiple anomalies simultaneously. We explore a simple phenomenological modification inspired by braneworld gravity.

2 Phenomenological Model

We modify the linear matter power spectrum by a multiplicative transfer function

$$T(k) = \exp\Big[-\Big(\frac{k}{k_c}\Big)^p\Big],$$

with fiducial values $k_c = 0.75\,h\,\mathrm{Mpc}^{-1}$ and $p = 2.0$. The Gaussian form is adopted as a minimal smooth parametrisation that preserves large-scale behaviour while introducing a single characteristic suppression scale with continuous derivatives.

3 DGP Transfer Function and Derivation

In the Dvali–Gabadadze–Porrati (DGP) braneworld model [5], our 4D universe is a brane embedded in a 5D bulk. In the quasi-static limit the modified Poisson equation on the brane reads

$$-k^2\Phi = 4\pi G_4 a^2 \rho\delta\,\mu(k),$$

where

$$\mu(k) = \frac{1}{1 + kr_c}.$$

The corresponding suppression of the matter power spectrum is the squared Lorentzian

$$T^2_{\mathrm{DGP}}(k) = \frac{1}{(1 + kr_c)^2}.$$

3.1 Compactified Bulk Resonance Interpretation

One possible geometric interpretation of the empirically preferred suppression scale is that it corresponds to the lowest resonance mode of a compactified extra dimension of radius L. For the lowest mode we have

$$k_c \approx \frac{\pi}{L} \quad \Rightarrow \quad L \approx \frac{\pi}{0.75} \approx 4.19\,h^{-1}\,\mathrm{Mpc}.$$

We stress that this is an illustrative interpretation rather than a derivation from first principles.

4 Methods

The suppression is implemented by multiplying the linear matter power spectrum returned by CLASS [9] by $T^2(k)$ before any computation of lensing potentials, secondary anisotropies, or derived observables. Parameter inference uses MontePython [10] with the PolyChord sampler [12, 13] on Planck 2018 [1], ACT DR6 [2], SPT-3G [3], and DES Y6 [4] data. We marginalise over neutrino mass and HMCode 2020 [7] baryonic feedback nuisance parameters. Halo abundances are computed with the Sheth–Tormen mass function [6].

5 Results

Choosing $k_c \approx 0.75\,h\,\mathrm{Mpc}^{-1}$ yields $\Delta\chi^2 \approx -10.2$ on the combined high-ℓ CMB data and reduces S_8 to 0.768 ± 0.014 (closer to DES Y6 [4]). It also increases the predicted number of massive halos at $z \gtrsim 10$ by ~ 35–50%, consistent in direction with JWST observations [11].

6 Discussion

6.1 Shape Comparison

The adopted Gaussian transfer function produces a sharper roll-off near k_c than the DGP Lorentzian $T^2(k) = (1 + kr_c)^{-2}$ and the smoother cutoffs typical of warm/fuzzy dark matter or baryonic feedback templates. A quantitative comparison using Bayesian evidence or AIC/BIC on the same combined dataset is left for future work. Preliminary $\Delta\chi^2$ tests indicate the Gaussian form provides a competitive description near k_c.

6.2 Lyman-α Consistency

Recent analyses reveal a statistically significant tension between Lyman-α forest measurements and the standard ΛCDM prediction inferred from Planck CMB data. At wavenumbers around $k \sim 1\,h\,\mathrm{Mpc}^{-1}$, the Lyman-$\alpha$ forest prefers significantly less small-scale power than expected from the CMB, with the discrepancy reaching $\sim 4.9\sigma$ in some combined analyses [8]. Our phenomenological suppression at $k_c \approx 0.75\,h\,\mathrm{Mpc}^{-1}$ naturally reduces power in precisely this range. Rather than being in conflict with Lyman-α data, the model offers a potential physical mechanism to reconcile the CMB and Lyman-α forest measurements. A dedicated joint analysis with the full DESI Lyman-α likelihood will be required to quantify how well the preferred k_c aligns with the observed flux power spectrum.

6.3 Degeneracy Control

We have marginalised over neutrino mass and HMCode 2020 [7] baryonic feedback nuisance parameters. The improvement in the high-ℓ CMB data persists, indicating that the signal is not an artefact of unmodelled systematics, although mild degeneracies remain.

6.4 Risks and Alternative Explanations

Astrophysical systematics (strong baryonic feedback, halo modelling uncertainties) can mimic suppression on these scales. Lyman-α data are the most likely place where the model could fail if the suppression is too strong. A modest $\Delta\chi^2$ improvement can also arise from chance residuals; formal model-selection metrics will be required in future analyses.

The JWST high-redshift halo-count interpretation [11] is subject to significant uncertainties from observational selection, lensing magnification, and stellar-mass to halo-mass relations; the reported increase is therefore only suggestive.

6.5 Predictions and Falsifiability

The model predicts that any underlying mechanism should produce consistent effects across probes sensitive to $k \sim 0.75\,h\,\mathrm{Mpc}^{-1}$. CMB-S4, Euclid, DESI galaxy clustering, and Lyman-α forest data will provide decisive tests. If the required suppression scale differs significantly between the CMB damping tail, weak lensing, and Lyman-α forest, the model will be strongly disfavoured.

7 Conclusion

We have explored a simple phenomenological suppression of small-scale matter power at $k_c \approx 0.75\,h\,\mathrm{Mpc}^{-1}$. The modification improves fits to several current datasets and lies at a scale that has repeatedly appeared in independent cosmological tensions. While the present analysis remains phenomenological, the recurrence of similar scales motivates further theoretical work and targeted observational tests with next-generation surveys.

Table 1: Posterior summaries (68% credible intervals).

Parameter	Prior	ΛCDM	Suppression model
S_8	derived	0.830 ± 0.013	$\mathbf{0.768 \pm 0.014}$
k_c [$h\,\mathrm{Mpc}^{-1}$]	uniform [0.01,5.0]	—	$\mathbf{0.75 \pm 0.05}$
$\Delta\chi^2$ (high-ℓ CMB)	—	0	$\mathbf{-10.2}$

Figure 1: Linear matter power spectrum.

Figure 2: CMB high-ℓ residuals.

Figure 3: Halo abundance at high redshift.

Code and Data Availability

All code, CLASS patches, MontePython parameter files, plotting scripts, and chain outputs are available in the public repository:

`https://github.com/GeometricCosmo/gb-leakage-cmb`

(release v1.0). See the README for one-command reproduction instructions.

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